SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE

PHASE 2: REQUIREMENT ANALYSIS PHASE

The Requirement Analysis Phase involved analyzing the requirements needed to implement secure and controlled record access in ServiceNow. The required table, fields, access conditions, user permissions, and expected system behavior were identified. These requirements were used as the foundation for designing and implementing the ACL.

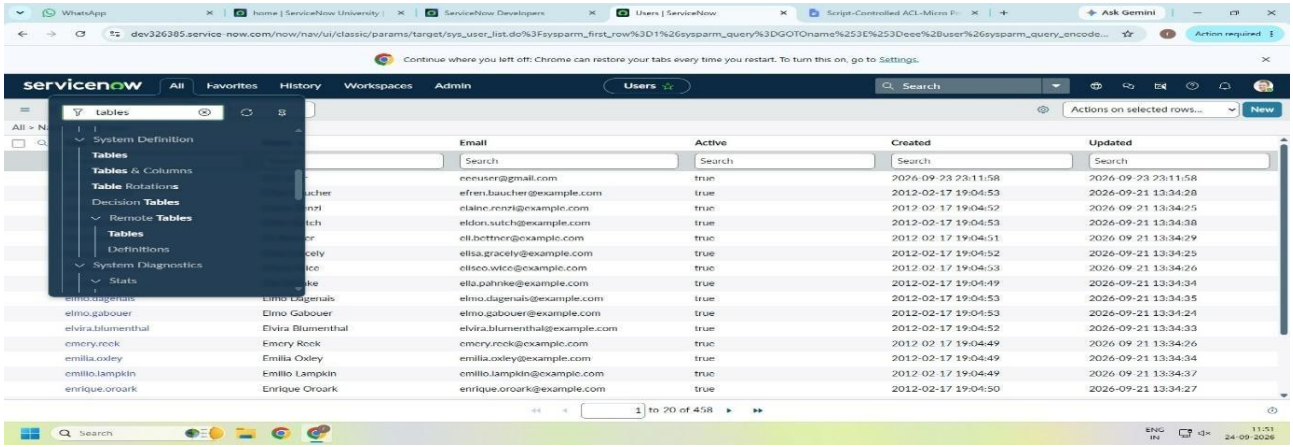
Before building anything, the exact ingredients of the solution were listed. The analysis answered four questions: which table holds the data, which fields decide who may see a record, under what condition access is allowed, and what a correct result will look like when the ACL is tested. The output of this phase is the list of requirements that the design in Phase 3 turns into an ACL.

2.1 Identifying the Required Table

The appropriate ServiceNow table was identified based on the project requirements. This table was selected as the target for implementing the required access control.

The data to be protected is stored in a new custom table named **Institution Details** (internal name *u_institution_details*). A separate table was created instead of using an existing one so that the ACL affects only this project data and does not interfere with the standard tables of the instance.

In the screenshot the Tables module under System Definition is opened from the application navigator. This is where a new table is created and where its columns are later maintained.

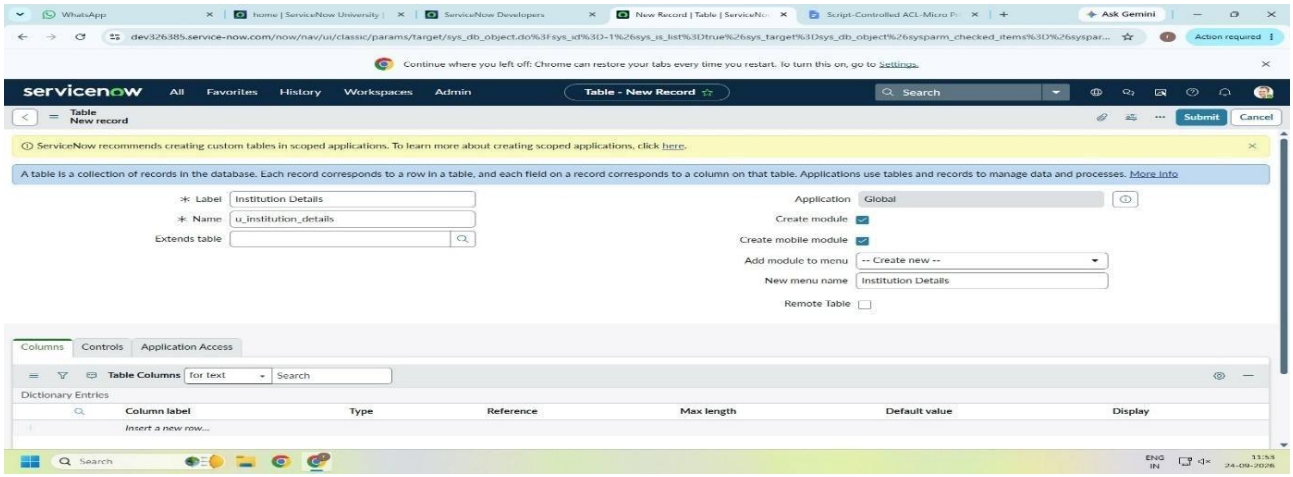


2.2 Identifying Required Fields

The important fields related to record access were identified and analyzed. These fields are used to define the conditions for controlling user permissions.

The table creation form asks for a label, a name, the table it extends, the application scope and options such as creating a module and adding a menu entry. Here the label is *Institution Details*, the name is generated as *u_institution_details*, the application is Global, and a module named Institution Details is created so that the table can be opened from the navigator.

The fields that hold the record data are then added as columns. Among them, **Branch** is the most important, because its value is the field that the ACL will use to decide whether a record can be seen.

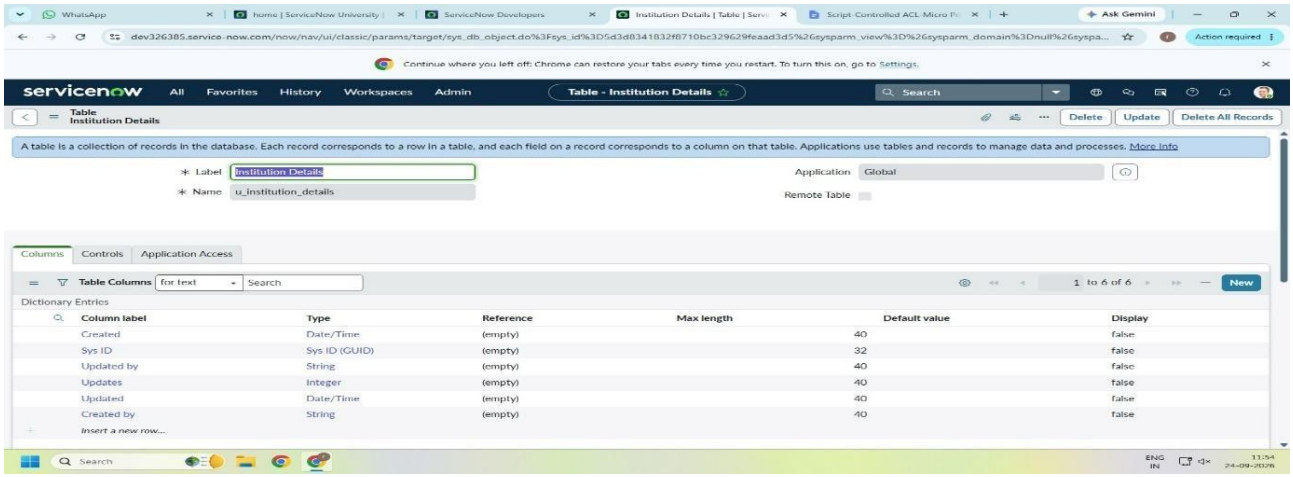


2.3 Defining Access Conditions

Specific conditions were defined to determine when a user should be allowed or restricted from accessing a record. These conditions form the basis of the ACL logic.

After the table was saved, its structure appeared with the standard system columns (Created, Sys ID, Updated by, Updates, Updated, Created by). These are added automatically to every ServiceNow table and are used for auditing.

The access condition was defined in terms of the Branch field: a record whose Branch is **EEE** is restricted to users who hold the branch role, and to administrators. Everyone else must be refused. Defining the condition on a field value, and not on a fixed list of record IDs, means that new records are protected automatically the moment they are created.



2.4 Analyzing Expected Access Behavior

The expected behavior of the access control was analyzed for different user and record conditions. This ensures that the configured ACL provides the required level of security.

The final column list of the Institution Details table contains the custom fields **Student Roll Number** (String), **Student Name** (Reference to User), **Faculty Name** (Reference to User), **Branch** (Choice), **Email** (String), **Phone Number** (String) and **Description** (String). Using reference fields for the student and faculty names links each record to real user accounts of the instance.

The expected behavior was written down as three cases. An administrator can see all records. A user with the branch role can see the EEE records. A user with neither of them sees an empty list. These three cases became the test cases used in Phase 6.

